

1. EcoTrack – Smart Campus Sustainability & Waste Management Platform

Industry Context

Universities generate large amounts of waste and consume significant water and energy every day. On a campus like Saveetha School of Engineering, issues such as overflowing bins, water leaks, and lights left on are usually reported through phone calls, word of mouth, or paper complaints. Reports are rarely tracked, and the maintenance team often lacks the exact location or photographic evidence of the problem. As a result, the campus faces:

- Low recycling rates (below 30%)
- Delayed resolution of reported issues
- Wastage of water and energy
- No visibility for the Green Campus Committee
- Poor student and staff participation
- Manual, scattered record keeping
- Missed sustainability targets

Current sustainability efforts are largely reactive rather than data-driven. Complaints are not logged centrally, reporters receive no updates, and management cannot measure progress toward zero-waste goals. There is a growing need for a mobile-first platform that lets every user report issues instantly, tracks their resolution, motivates participation, and gives decision-makers real-time insights.

Problem Statement

Design and develop EcoTrack, a mobile-first web platform that enables students and faculty to report waste, water, and energy issues on campus with photo and location tagging.

The platform should allow users to track report status, notify them of progress, and let maintenance staff receive and resolve issues by priority and assigned zone.

Using role-based access, the system should provide the Green Campus Committee with a real-time dashboard, filters, and automated monthly sustainability reports, while encouraging participation through points, badges, and leaderboards.

The solution should be developed using Agile Scrum, remain scalable and secure, and be designed for future integration of IoT sensors such as smart bins and water meters.

Objectives

The platform should:

- Enable issue reporting with photo and GPS location
- Track report status from submission to resolution
- Send push and in-app status notifications
- Provide a real-time committee dashboard with map view
- Gamify participation with points, badges, and leaderboards
- Generate automated monthly sustainability reports
- Manage user roles and maintenance zone assignments

- Support future IoT sensor integration

Expected Outcomes

The proposed platform will:

- Increase the campus recycling rate to 60% or more.
- Speed up resolution of waste, water, and energy issues.
- Give the committee real-time, data-driven visibility.
- Improve student and staff engagement through gamification.
- Reduce water and energy wastage through quick reporting.
- Support management with automated sustainability reports.
- Provide a scalable, IoT-ready base for a zero-waste campus.